

EXERCISE SET 2.2

- 13.** Use truth tables to verify the following logical equivalences. Include a few words of explanation with your answers.

a. $p \rightarrow q \equiv \sim p \vee q$

b. $\sim(p \rightarrow q) \equiv p \wedge \sim q$.

a. $p \rightarrow q \equiv \sim p \vee q$

p	$\sim p$	q	$p \rightarrow q$	$\sim p \vee q$
T	F	T	T	T
T	F	F	F	F
F	T	T	T	T
F	T	F	T	T

First, I decided to break down the statement variables p , q as well as $\sim p$ into their possible states, before taking on the compound statements individually.

Upon comparison, we can conclude that $p \rightarrow q$ and $\sim p \vee q$ are logically equivalent, because they both share the same truth set.

Solution: **a.** Yes, $p \rightarrow q$ and $\sim p \vee q$ are logically equivalent: $p \rightarrow q \equiv \sim p \vee q$

b. $\sim(p \rightarrow q) \equiv p \wedge \sim q$.

p	q	$\sim q$	$p \rightarrow q$	$\sim(p \rightarrow q)$	$p \wedge \sim q$
T	T	F	T	F	F
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	F	F

Again, I decided to break down the statement variables p , q and $\sim q$ into their possible states, before taking on the statement forms individually.

Additionally, I also decided to note down all possible values of $p \rightarrow q$, which made it easier to map the negated values for $\sim(p \rightarrow q)$. This is not a required step.

We can then wrap up the table with possible values for $p \wedge \sim q$, and compare the results.

We can also conclude in this case that $\sim(p \rightarrow q)$ and $p \wedge \sim q$ are logically equivalent, because they both share the same truth set.

Solution: **b.** Yes, $\sim(p \rightarrow q)$ and $p \wedge \sim q$ logically equivalent: $\sim(p \rightarrow q) \equiv p \wedge \sim q$